

AC IONISING EQUIPMENT



What is AC Static Elimination?

A normal mains 110/240V, 50/60Hz AC supply is increased by a special transformer to typically 7kV AC. This high voltage is carried by a shielded HV cable to the static eliminator, where it is connected to an array of emitter pins. This creates a high energy ion cloud, in which a

very large number of positive and negative ions are generated. As the AC cycle changes, either positive or negative ions are produced in approximately equal quantities. A statically charged surface of either polarity passing close to this ion cloud will be quickly neutralised.

AC Applications

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Popular Application Example	Product																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
		Food and Drinks			Rigid Plastics		Flexible Plastics		Printing			Converting		Pharma / Medical		Other					Drinks					Food		Product Packaging		Product Handling		Thermoforming		Plastic Rigid		Blow Moulding		Plastics Film Extrusion		Plastic Film Converting		Printing Sheet		Printing Web		Printing Other		Print Finishing		Converting Paper		Textiles		Bag Making		Medical		Pharmaceutical		Cosmetics		Electronics		Clean Room		Automotive		Metal/Glass Processing																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
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AC Power Units

Meech AC power supplies provide between a 3.5 and 7kV and are compatible with all Meech AC ionising equipment. Illuminated indicators allow easy monitoring of the units.

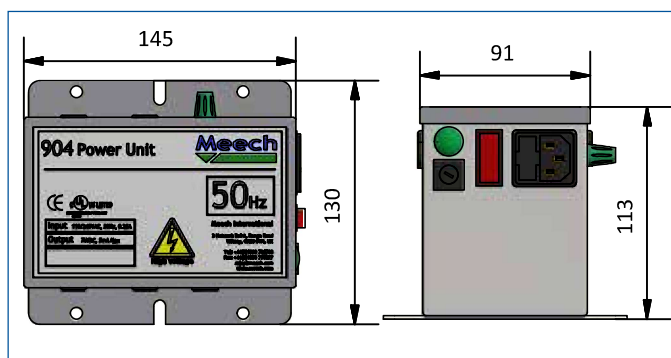
	904 Power Supply		905 Power Supply		Hyperion 906 Power Supply		Hyperion 907 Power Supply	
								
Input Voltage	110/240V AC				24V DC			
Output Voltage	7kV		5kV		Adjustable, 4-7kV			
Fault alert	Yes		Yes		Yes		Yes	
Cleaning alert	No		No		Yes		Yes	
Spark alert	No		No		No		Yes	
Dimensions	145 x 130 x 113mm				95 x 118 x 110mm			
Weight	3.4kg				2.2kg			
Output Frequency	50/60Hz				67Hz			
Input Frequency	50/60Hz							
Output Current	<5mA							
Max Input Current	1.3A				850mA			
Connection	IEC mains connector				Murr 4 pin M8, Earth terminal			
Output signal	Via Meech 900vs2				24V, Output Resistance 10kΩ			
Settings control	n/a				BarMaster			
Indicator	1 x HT LED				2 x Green/ red LEDs			
Scan QR Code for Operating Manual								



904 & 905 Power Supply

The 904 and 905 Power Supplies are constant voltage power supplies, designed to provide a 7kV (904) or 5kV (905) source for Meech AC ionising equipment. They are adjustable to operate at 100-120V or 200-240V input voltage and are available in either 50Hz or 60Hz versions to suit local conditions. They both incorporate a low voltage jack socket for connection of the 900vs2 High Voltage Sensor.

- Multiple connection ports that allow up to 4 ionisers to be connected to 1 power unit.
- Modified output wave form that allows more powerful neutralisation, lower electrical stress and increased reliability.
- Illuminated indicator for high voltage output, shows the system is operational and can also assist in ionising system diagnosis.
- Mains voltage selector switch allows easy selection of mains voltage.



HYPERION⁺

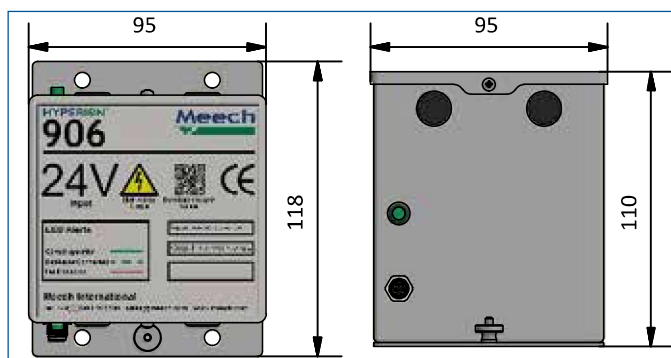
A Meech Innovation

Hyperion 906 & 907 Power Supply

The Hyperion 906 & 907 are advanced AC power supplies. Compatible with all Meech AC Ionisers the 906 & 907 are powered by 24V DC and produce an adjustable output of up to 7kV AC.

Each unit will provide a high voltage connection for up to four ionisers. Ion Current Monitoring provides the operator with local and remote alerts when the ionising equipment needs cleaning.

- Universal 24V input voltage.
- Output voltage can be adjusted between 4 and 7kV.
- Easy to see LED indicators.
- HT failure Alert shows local and remote alert of damage or HT failure.
- Clean Pin alert gives local and remote instruction to clean the bar (Hyperion 907 only).
- Spark detection system will detect any problem as it happens (Hyperion 907 only).
- Balance adjustment allows AC ionisers (for the first time) to be had on sensitive electronics applications (Hyperion 907 only).



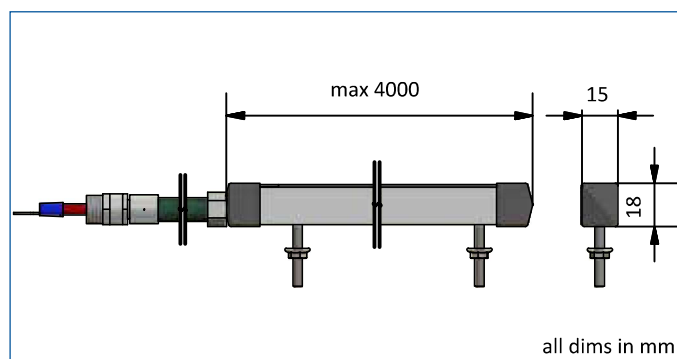
AC Ionising Bars

Meech AC Ionising Bars are powered by external AC Power supplies. They offer 4.5 to 7kV AC output for short range ionisation. The resistive coupling makes them shockless and safe for operators.

	914 Anti-Static Bar	915 Anti-Static Bar	910 Anti-Static Bar	912 Anti-Static Bar
				
Dimensions (W x H)	15mm x 18mm	18mm x 42mm	16mm x 20mm	21mm x 18mm
Max. Length	4000mm	4000mm	4000mm	Range from 30mm to 254mm internal dia
Weight	400g/ metre	1100g/ metre	750g/ metre	350g for 30mm internal dia
Max. Temperature	85°C	60°C	85°C	85°C
Operating voltage	5.0kV AC	4.5-7.0kV AC	7.0kV AC	7.0kV AC
Operating current	< 5mA			
Cable	2000mm			
Construction	Anodised aluminium and PVC			
Emitters	Titanium		Stainless steel	
Mounting	M4 x 20mm studs	M4 x 20mm studs	M4 x 20mm studs	n/a
Optimum Working Distance	25-100mm	30-150mm	15-30mm	15-30mm
Scan QR Code for Operating Manual				

914 Ionising Bar

The 914 AC Ionising Bar has been designed to meet the most arduous of static elimination problems, including those encountered in high speed web applications. The powerful performance provides very fast decay times and effective ionisation up to distances of 100mm. The 914 Ionising Bar is powered by the 905 Power Supply.

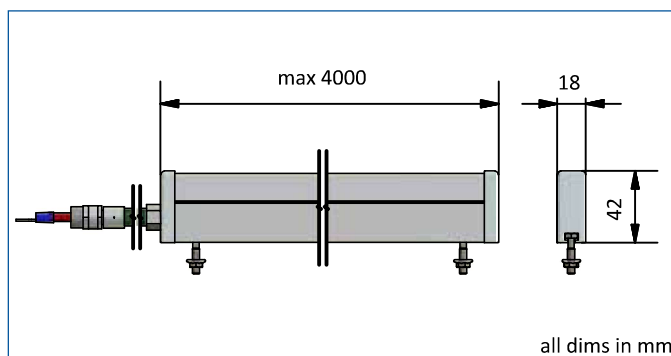


- Powerful and intense ion production leads to fast decay times.
- Slim profile means the 914 bar can fit where space is limited.
- Resistively coupled emitter pins ensures the 914 bar is completely shockless.
- Sharp titanium emitter pins reduce pin contamination and extend life of bar.

915 Ionising Bar

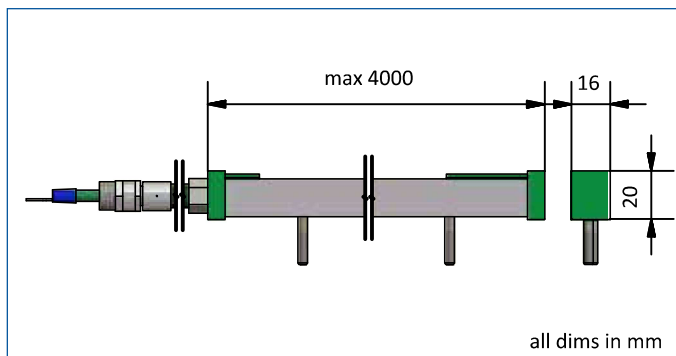
The 915 AC Ionising Bar has been designed to meet the most arduous of static elimination problems, including those encountered in high speed web applications. The powerful performance provides very fast decay times and effective ionisation up to distances of 150mm. The 915 Ionising Bar is powered by the 904 Power Supply.

- Powerful and intense ion production leads to fast decay times.
- Extended range performance offers flexibility in positioning.
- Slim profile means the 915 bar can fit where space is limited.
- Resistively coupled emitter pins ensures the 915 is completely shockless.
- Sharp titanium emitter pins reduce pin contamination and extend life of bar.
- The 915 bar is designed to be quick and easy to maintain.



910 Ionising Bar

The 910 Ionising bar incorporates an inductively coupled design that achieves powerful static neutralisation with completely shockproof operation. The design is compact, yet rugged enough for most industrial applications. The 910 Ionising Bar is powered by the 904 Power Supply.



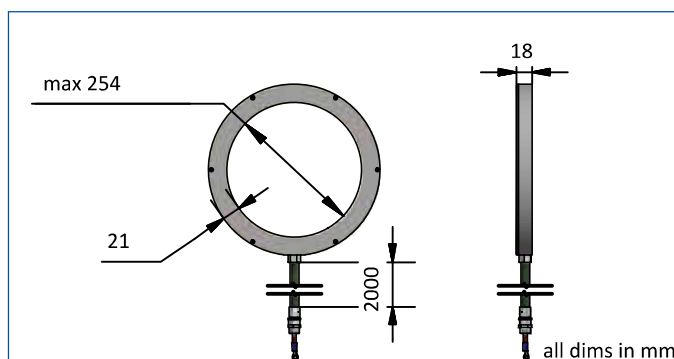
- Powerful and intense ion production leads to fast decay times making it perfect for fast web speeds.
- Slim profile means the 910 bar can fit where space is limited.
- Short-circuit proof, bar will still perform even when one electrode is accidentally earthed.
- Sealed versions available for use in harsh environments.

912 Ionising Bar

The 912 circular static eliminator is based on the popular 910 Ionising Bar. The inductively coupled design provides powerful static neutralisation and completely shockproof operation. The 912 Ionising Bar is powered by the 904 Power Supply and is suitable for applications where 360° neutralisation is required.



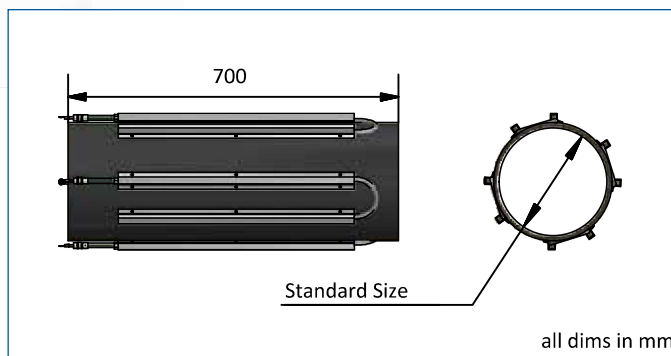
- Suitable for applications that requires complete 360° ionisation.
- Powerful and intense ion production leads to fast decay times.
- Short-circuit proof, bar will still perform even when one electrode is accidentally earthed.



913 Flow Tube

The 913 Flow Tube neutralises static charges on materials in handling and conveying systems. It incorporates in-line ionisation, which eliminates material clinging and clogging in ducting systems due to static charges. The 913 Flow Tube is powered by 905 power supply and is provided in a range of diameters to fit most of the industrial standard installations.

- Available in a range of diameters to easily incorporate into existing ducting.
- Highly effective neutralisation of static charges
Allows product to flow freely, thereby preventing clogging and improving productivity.

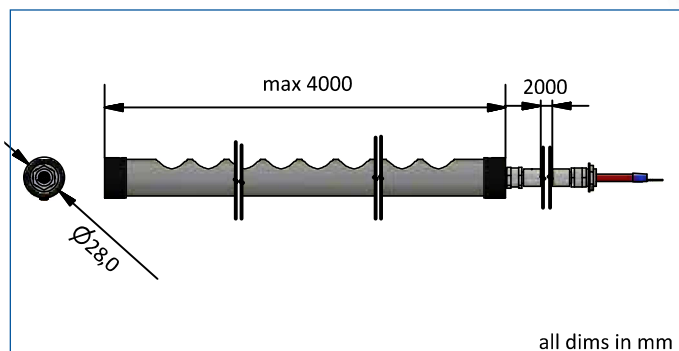
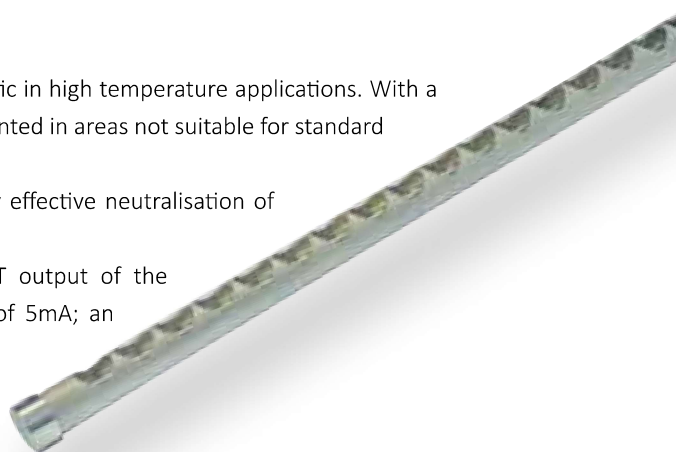


925 High Temperature Bar

The Meech 925 is an AC Ionising bar, designed to eliminate static in high temperature applications. With a maximum operating temperature of 150°C, the 925 can be mounted in areas not suitable for standard AC ionising bars.

The powerful performance of this anti-static bar provides very effective neutralisation of static charges at distances up to 100mm.

The emitter pins of the 925 are directly coupled to the HT output of the 905 power supply. The 905 has a maximum output current of 5mA; an internationally recognised safe current.



- Very fast decay times makes the 925 effective for fast web speeds.
- To ease maintenance and increase service life, the 925 has a detachable cable assembly. This allows the bar to be removed from the machine for cleaning operations on installations that make in-situ cleaning difficult.
- Slim profile means the 925 can be fitted in restricted spaces.

AC Ionising Blowers & Air Curtains

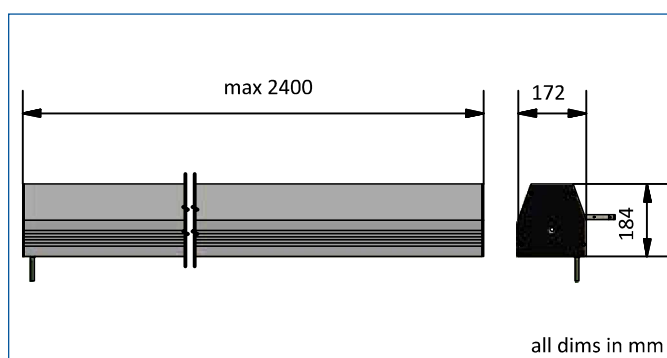
Meech AC Ionising Blowers and Air Curtains provide ionisation over longer and wider areas. Ionisation is boosted by either a fan or compressed air system.

	935 Ionising Blower	957 Ionising Air Curtain
		
Dimensions	172 x 184 x up to 2400mm	28 x 62.8 x up to 1800mm
Weight	4.5kg per 400mm length	2.65kg/1m length
Input Voltage	110/240V AC	
Output Voltage	7kV	
Input Frequency	50/60Hz	
Short Circuit Current	<5mA	
Compressed Air consumption	n/a	43cfm(122l/min)@80psi / 1000mm 5.4 Bar per 25mm
Compressed air connection	n/a	1/4" BSP female
Max. Temperature	60°C	85°C
Mounting	Steel fixing brackets supplied with each unit	M4 Stud
Cable length	3000mm	2000mm
Connection	IEC mains connector	Crimp terminal / 1/4" BSP
Settings control	Fan speed	n/a
Optimum Working Distance	200- 1500mm	50-300mm
Scan QR Code for Operating Manual		

935 Ionising Blower

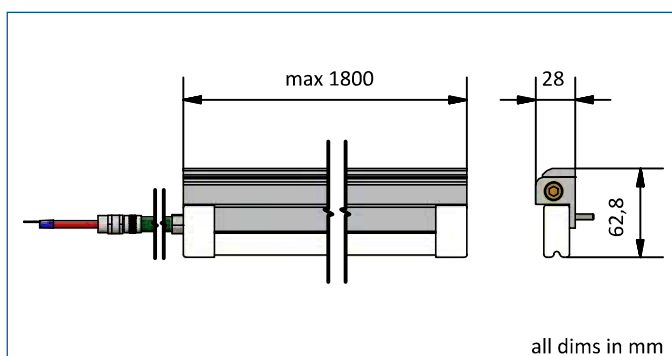
The 935 Ionising Blower provides effective long range ionisation. The Blower comprises an ionisation head and an integrated fan system. A high volume flow of ionised air is generated by blowing air through the ionisation head. Ionisation is provided by powerful 915 Ionising Bars and powered by the 904 Power Supply.

- Available in 110V and 240V versions.
- Resistively coupled ionisation head for shockless operation.
- High volume air flow provides effective long range ionisation.
- Available in 200mm length increments for wide range of applications.
- Variable fan speed (option) to suit specific application.



957 Ionising Air Curtain

The 957 Ionising Air Curtain provides effective static neutralisation and dust removal in a wide range of industrial applications. The unit comprises a powerful Model 915 ionising bar mounted on to an Energy Saving Air Curtain. The 957 Ionising Air Curtain requires a compressed air supply and is powered by the 904 Power Supply.



- Laminar air flow reduces recontamination and effective range.
- Air Curtain removes dust contamination and reduces air consumption.
- Easy maintenance to optimise performance and increase equipment life.

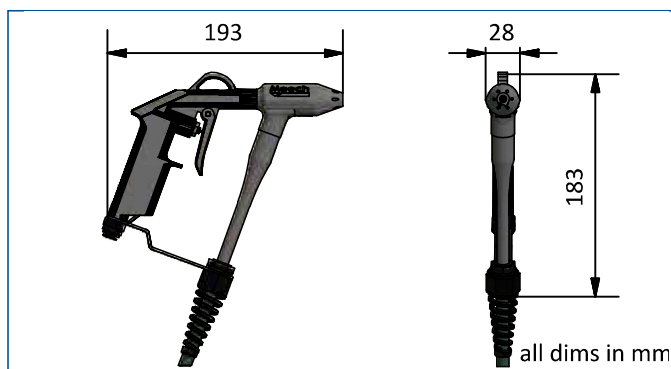
AC Ionising Guns & Nozzles

Meech AC Ionising Guns and Nozzles use compressed air to offer dust removal as well as neutralising any static charge.

	954v2 Ionising Gun	954v2 Ionising Nozzle	940 Ionising Nozzle	942 Ionising Nozzle	942F Flexi Nozzle
					
Dimensions	193 x 183 x 28mm	71 x 26 x 28mm	56 x 29 x 25mm	32 x 32 x 55mm	32 x 32 x 55mm Trunking (standard) - 118mm long
Output Voltage	7kV	7kV	5kV	5kV	5kV
Weight	490g	215g	200g	230g	240g
Input Frequency	50/60Hz				
Input Voltage	110/240V AC				
Cable length	3 metres of H.T. cable in flexible conduit as standard	2 metres of H.T. cable in flexible conduit as standard	made to order	3 metres of H.T. cable in flexible conduit as standard	3 metres of H.T. cable in flexible conduit as standard
Compressed Air Consumption	4cfm at 1 bar	5.5cfm at 1 bar	4cfm at 1 bar	9cfm at 1 bar	9cfm at 1 bar
Compressed air connection	6mm push-in fitting	1/4" BSP	1/8"	1/2"	1/2"
Short Circuit Current	<5mA				
Max. Temperature	85°C				
Optimum Working Distance	up to 500mm				
Scan QR Code for Operating Manual					

954v2 Ionising Gun

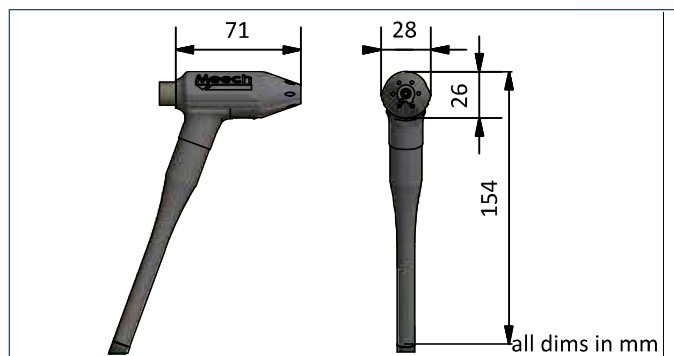
The 954v2 Ionising Gun is the latest evolution of hand held air assisted ionising guns from Meech. Designed to neutralise static charges and remove dust contamination, the 954v2 has decay times 60% faster than the previous model.



- Unique nozzle design offers more effective ionisation performance and reduced noise levels.
- Resistively coupled ionisation head for shockless operation.
- 945v2 meets dead-end pressure safety guidelines.
- Ergonomic, lightweight design making it comfortable for prolonged use.
- Decay rates 60% faster than the previous 954 Ionising Gun.

954v2 Shockless Ionising Nozzle

The 954v2 Nozzle is designed to neutralise static charges and remove dust contamination. It includes a 1/4" BSP threaded fitting for easy connection to any air supply.



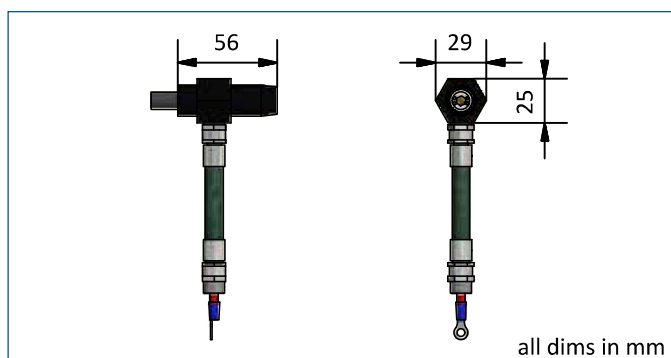
- Unique nozzle design offers more effective ionisation performance and reduced noise levels.
- Resistively coupled ionisation head for shockless operation.
- 945v2 meets dead-end pressure safety guidelines.
- Decay rates 60% faster than the previous 954 Ionising Nozzle.

940 Ionising Nozzle

The 940 Ionising Nozzle is available as a single unit (Blank End) or with two or more units connected in series (Through Type). Powered by a 905 Power Supply the 940 Nozzle is suitable for applications requiring the removal of electrostatically attracted dust and for longer range static neutralisation.

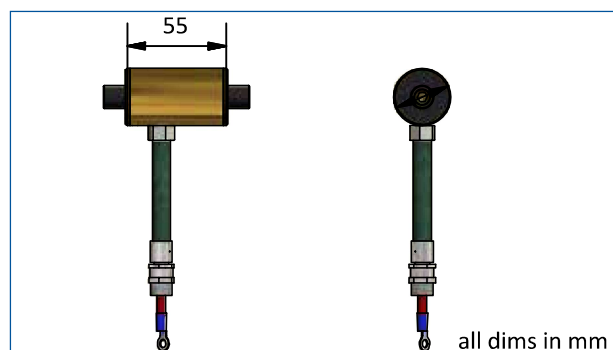


- Easy maintenance with minimum downtime.
- Air consumption of 16cfm (453 l/min) @ 100 psi (6.9 Bar).
- Small and compact so will easily fit into confined spaces.



942 Ionising Nozzle & 942F Flexi Nozzle

The 942 is a small, lightweight in-line ionising air nozzle. It can be used either as a single unit (Blank End) or with two or more units connected in series (Through Type). The 942F includes a length of flexible knuckle trunking. They are particularly suited to applications on sheet feeder and delivery systems. The 942 and 942F are powered by the 905 Power Supply.



- Highly efficient air flow characteristics.
- Rapid decay times.
- Easy maintenance with minimum downtime.
- Small and compact so will easily fit into confined spaces.



Options for AC Equipment



900vs2 High Voltage Sensor

The 900vs2 is a high voltage sensor designed to monitor the high voltage output of Meech AC power units. Operating status is displayed via LED's. Relay outputs allow connection to remote indicator lamps or alarms. The 900vs2 is powered directly from the 904 or 905 Power supply.



Dust Proof for 915 Bars

For installation in dirty or dusty environments the dust proof option improves resilience of the bar.



Water Resistant

A water resistant option allows the installation of bars where routine machine cleaning can cause them to be splashed.



Air Assist

The effective range of bars can be increased by the addition of an air-boost system. The air assist option is suitable for 914 or 915 ionising bars.



900-HTCON In-line Connector for AC Bars

On machinery where AC bars need to be disconnected periodically, the quick connect options provides a simple solution. The connector can be used to provide a quick disconnect system for ionising bars or can be used to overcome problematic bar installations where machine bulk heads make cable routing difficult.